France Top 50 Playlist Analysis

Audience Sensitivity, Explicit Content, Chart Performance, and Format Preferences

Project Report | Prepared for finalization with verified dataset results

GitHub repository

https://github.com/Ishan2025-new/Unified-Mentor-Project-1-France-Top-50-Playlist-Analysis

Live Streamlit dashboard

https://france-top-50-playlist-2026.streamlit.app/

Abstract

This project examines a France Top 50 playlist dataset through descriptive analysis of explicit-content labels, chart position, popularity, song duration, album type, artist-level patterns, and changes over time where date information is available. The goal is to organize the dataset into interpretable summaries and provide an interactive Streamlit dashboard for exploration.

This report describes the project design and analysis plan. Dataset-specific numerical findings are intentionally not stated here because the underlying cleaned dataset and calculated outputs were not supplied with this report-generation request. Before final submission, replace the marked results prompts with verified values and charts from the notebook or deployed app.

1. Introduction

Music playlist analysis can help describe which tracks appear in a chart snapshot, how tracks differ in duration and release format, and how metadata labels are distributed across the playlist. Chart position and popularity are useful descriptive variables, but they should not be interpreted as proof that a particular track feature causes success.

This project focuses on four connected areas: audience-sensitivity indicators, explicit-content metadata, chart and popularity comparisons, and format characteristics such as duration and album type.

2. Objectives

• Summarize the dataset using track and artist counts and available popularity and duration measures.

• Describe the distribution of explicit and non-explicit labels, including records with missing or unrecognized labels.

• Compare explicit-content labels across chart-position groups, popularity groups, album types, and artists.

• Examine song-duration and album-type distributions and compare them with chart position and popularity.

• Explore temporal patterns only when valid dates or repeated snapshots are available.

• Present the analysis in an interactive Streamlit application and communicate limitations responsibly.

3. Dataset and Variables

The analysis uses the project's France Top 50 playlist dataset. The dashboard expects the following fields, or recognized equivalents: song, artist, is_explicit, date, position, popularity, duration_min, and album_type.

Song and artist identify the track and performer. The explicit field is treated as supplied metadata. Date represents the playlist snapshot or chart date. Position represents chart rank, where a smaller number generally indicates a higher position. Popularity is interpreted according to the source platform's scale. Duration is expressed in minutes. Album type represents the release category recorded in the source data.

Before submission, document the source of the dataset, collection date or date range, number of rows, number of unique tracks and artists, and any exclusions made during cleaning.

4. Methodology

4.1 Data preparation: standardize column names, convert numeric fields to numeric types, parse dates, normalize explicit labels to true/false where possible, and preserve unknown labels as missing rather than guessing their meaning.

4.2 Descriptive statistics: calculate record counts, unique-song and unique-artist counts, mean and median duration where useful, average popularity, and explicit-label proportions among records with valid labels.

4.3 Chart-position analysis: group records into Top 10, positions 11–25, and positions 26–50 when the dataset supports these ranges. Compare the count and proportion of explicit labels in each group.

4.4 Popularity analysis: compare explicit-label distributions across popularity bands. Confirm the popularity scale and bin definitions match the source data.

4.5 Format analysis: group duration into clearly documented intervals and summarize album types. Compare mean or median popularity and chart position by duration group and album type.

4.6 Artist-level analysis: calculate each artist's number of records and explicit-label share using only valid labels. Interpret small sample sizes cautiously.

4.7 Time analysis: plot trends only when date values represent meaningful repeated observations. A single snapshot cannot establish a trend over time.

4.8 Interactive presentation: use Streamlit controls to filter by artist, album type, and explicit label, and provide charts, tables, and filtered-data export.

5. Analysis Plan and Results to Complete

Important: the following are reporting prompts, not measured findings. Populate them from the actual notebook outputs or the live dashboard before submitting the report.

5.1 Dataset overview — Report the number of playlist records, unique songs, unique artists, date range, and missing-value counts.

5.2 Explicit-content distribution — Report the number and percentage of explicit, non-explicit, and unknown labels. State the denominator used for percentages.

5.3 Chart position by explicit status — Compare both counts and within-group percentages across chart-position groups. Avoid claiming that explicit status determines ranking.

5.4 Song duration by explicit status — Report mean and median duration for each label group, sample sizes, and any notable difference supported by the data.

5.5 Album-type distribution — Report counts and proportions for each album type, including unknown categories if present.

5.6 Duration by album type — Compare duration distributions by album type and include sample sizes to make the comparison interpretable.

5.7 Format patterns by chart position and popularity — Compare duration and album type across rank and popularity groups. Describe associations, not causal effects.

5.8 Artist-level patterns — Identify artists with enough observations for a fair comparison and report each group's denominator.

5.9 Trends over time — Include only if multiple dates or snapshots exist. Describe the date coverage and whether counts represent unique tracks or playlist appearances.

6. Streamlit Application

The deployed dashboard is available at: https://france-top-50-playlist-2026.streamlit.app/

The source-code repository is available at:
https://github.com/Ishan2025-new/Unified-Mentor-Project-1-France-Top-50-Playlist-Analysis

The application is organized into Overview, Audience Sensitivity, Content Compliance, Format Preferences, and Data Explorer sections. Filters allow users to narrow results by artist, album type, and explicit-content status. The Data Explorer supports searching and downloading the filtered records as CSV.

Before final submission, open the deployed app in a fresh browser session and confirm that the dataset loads, each section renders without errors, filters update the charts, and the CSV download works.

7. Discussion

The dashboard is intended as a descriptive exploration tool. Differences between explicit and non-explicit groups may be associated with other factors, including artist, genre, release strategy, playlist selection, label practices, and platform-specific popularity calculations. The analysis does not isolate these factors unless a separate statistical design is added.

An explicit-content label is not a complete measure of audience sensitivity. Listeners differ in preferences and context, and platform metadata may be incomplete or inconsistent. Similarly, the presence of a track in a Top 50 playlist is not evidence that all listeners prefer its content or format.

8. Limitations and Ethics

• Findings depend on the coverage, accuracy, and collection date of the supplied dataset.

• Missing, duplicated, or inconsistently labelled records can affect summary statistics.

• A single chart snapshot cannot support claims about trends over time.

• Observational comparisons do not establish causation.

• Artist-level comparisons can be misleading when sample sizes are small.

• Explicit metadata should not be treated as a legal determination or a certification of platform compliance.

• The report should avoid making claims about listener demographics or audience reactions unless those data were directly collected and analyzed.

9. Conclusions and Recommendations

The project provides a structured workflow for examining explicit-content metadata and format characteristics within a France Top 50 playlist dataset. Its main deliverable is an interactive dashboard that brings together summary metrics, comparisons by chart position and popularity, format analysis, and data exploration.

Recommendations: (1) validate metadata and document missing values before analysis; (2) report counts alongside percentages; (3) compare both average and median duration when distributions are skewed; (4) avoid overinterpreting artists represented by very few records; (5) use repeated snapshots for temporal conclusions; and (6) keep conclusions descriptive unless an appropriate inferential design supports stronger claims.

10. Final Submission Checklist

■ Add dataset source, extraction date, row count, unique-track count, and unique-artist count.

■ Replace every results prompt with verified numerical findings from the actual dataset.

■ Insert final charts and label axes, units, groups, and sample sizes.

■ Confirm the deployed app works and the repository contains the app code and requirements.

■ Proofread the report and ensure every statement is supported by the analysis.

■ Add references for the dataset, platform documentation, and any external methods used.

Appendix A: Project Links

GitHub repository: https://github.com/Ishan2025-new/Unified-Mentor-Project-1-France-Top-50-Playlist-Analysis

Deployed Streamlit application: https://france-top-50-playlist-2026.streamlit.app/